Exp 2

Aim:

Study of different types of network cables.

a) Understand different types of network cable.

1) Unshielded twisted pair (UTP)
2) Shielded Twisted Pair (STP)
3) Coaxial Cable
4) Fibre optic cable

Cable type	Category	Maximum Data Transmission	Advantage/ Disadvantage	Application/ use
UTP	Category 3	10 bps	Advantages:- ⇒ Cheaper in cost ⇒ Easy to install as they have a smaller diameter	10 Base-T Ethernet
	Category 5	Upto 100 Mbps	Disadvantages:- ⇒ More prone to (EMI) Electromagnetic inference and noise	Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: ⇒ Shielded ⇒ Faster than UDP ⇒ Less susceptible to noise & inference	Gigabit Ethernet 10G Ethernet (55m) widely used in data centers
SSTP	Category 7	10 Gbps	Disadvantages: ⇒ Expensive ⇒ Greater Installation effort	Gigabit Ethernet 10G Ethernet (100 m)

Coaxial Cable	RG1 - 6 RG1 - 59 RG1 - 11	10-100 Mbps	⇒ High bandwidth ⇒ Immune to interfer-ence ⇒ Low loss bandwidth ⇒ Versatile Disadvantages: Limited distance, costly Size is bulky	Speed of is 500 m Television network Higher Speed internally Connections.
fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages: ⇒ High speed ⇒ High bandwidth ⇒ High security ⇒ Loss distance Disadvantages: ⇒ Expensive ⇒ Requires skilled installers	Maximum distance of fibre optics Cable is around 100 meters.

Result:

Thus different types network Cables were studied.